Adaptive Mesh Refinement for Supersonic Flow

Mach 3 Cylinder at 50 km Altitude

OpenFOAM v2406 | Gradient-Based Adaptation | Unstructured Gmsh Mesh

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Problem Statement & Motivation

Problem

Supersonic flows produce strong shock waves with steep gradients

Uniform meshes waste cells in smooth regions far from shocks

Fine uniform meshes are computationally expensive

Need automated refinement that focuses cells where needed

Objective

Develop automated AMR pipeline for unstructured meshes

Use pressure gradient as adaptation criterion

Demonstrate on Mach 3 cylinder at 50 km altitude

Pipeline: Gmsh + OpenFOAM + ParaView (fully automated)

Why This Matters

Supersonic/hypersonic vehicle design requires accurate shock resolution without excessive computational cost

Gradient-based adaptation automatically concentrates cells at bow shocks, wake regions, and stagnation points

The pipeline is solver-agnostic: works with OpenFOAM (CFD) and COMET (DSMC) through a unified input file

Single configuration file (amr_pipeline.input) controls the entire process; no script editing required

Flow Conditions

1976 US Standard Atmosphere at 50 km Altitude

Parameter	Value	Unit
Altitude	50	km
Mach Number	3.0	-
Freestream Velocity	990	m/s
Freestream Pressure	79.78	Pa
Freestream Temperature	270.65	K
Freestream Density	1.027 x 10^-3	kg/m^3
Speed of Sound	330	m/s
Wall Temperature	800	K
Turbulence Model	Laminar	-

Notes

Density computed: rho = p / (R * T) = 79.78 / (287.05 * 270.65)

Speed of sound: a = sqrt(gamma * R * T) = sqrt(1.4 * 287.05 * 270.65) = 330 m/s

Mach number: M = V/a = 990/330 = 3.0

Reference: NASA-TM-X-74335 (1976 US Standard Atmosphere)

Domain Geometry

Rectangular domain: 1.0 m x 0.8 m

Cylinder diameter: 0.2 m, centered at (0.5, 0.4)

Quasi-2D: single cell in z-direction (0.01 m extrude)

Unstructured triangular mesh generated by Gmsh

Computational Setup & Boundary Conditions

Boundary	Velocity (U)	Pressure (p)	Temperature (T)
Inlet (left)	fixedValue: (990, 0, 0) m/s	fixedValue: 79.78 Pa	fixedValue: 270.65 K
Outlet (bottom, right, top)	zeroGradient	zeroGradient	zeroGradient
Wall (cylinder)	noSlip	zeroGradient	fixedValue: 800 K
Front & Back	empty	empty	empty

Initial Conditions

Internal field initialized to freestream values

U = (990, 0, 0) m/s everywhere

p = 79.78 Pa everywhere

T = 270.65 K everywhere

Simulation starts from uniform initial state

Gas Properties (thermophysicalProperties)

Equation of state: Perfect gas (hePsiThermo)

Molecular weight: 28.9 g/mol (air)

Specific heat: Cp = 1005 J/(kg K)

Transport: Sutherland law (As=1.458e-6, Ts=110.4 K)

gamma = 1.4 (ratio of specific heats)

Solver & Numerical Schemes

Solver: rhoCentralFoam

- Density-based compressible solver
- Kurganov-Tadmor central scheme for flux evaluation
- Semi-discrete, non-staggered formulation
- Suitable for high-speed flows with strong shocks
- No Riemann solver needed (central differencing)

Reconstruction & Limiters

- Density (rho): Minmod limiter
- Velocity (U): MinmodV limiter (vector)
- Temperature (T): Minmod limiter
- Minmod is most diffusive but most stable for strong shocks
- Prevents spurious oscillations near discontinuities

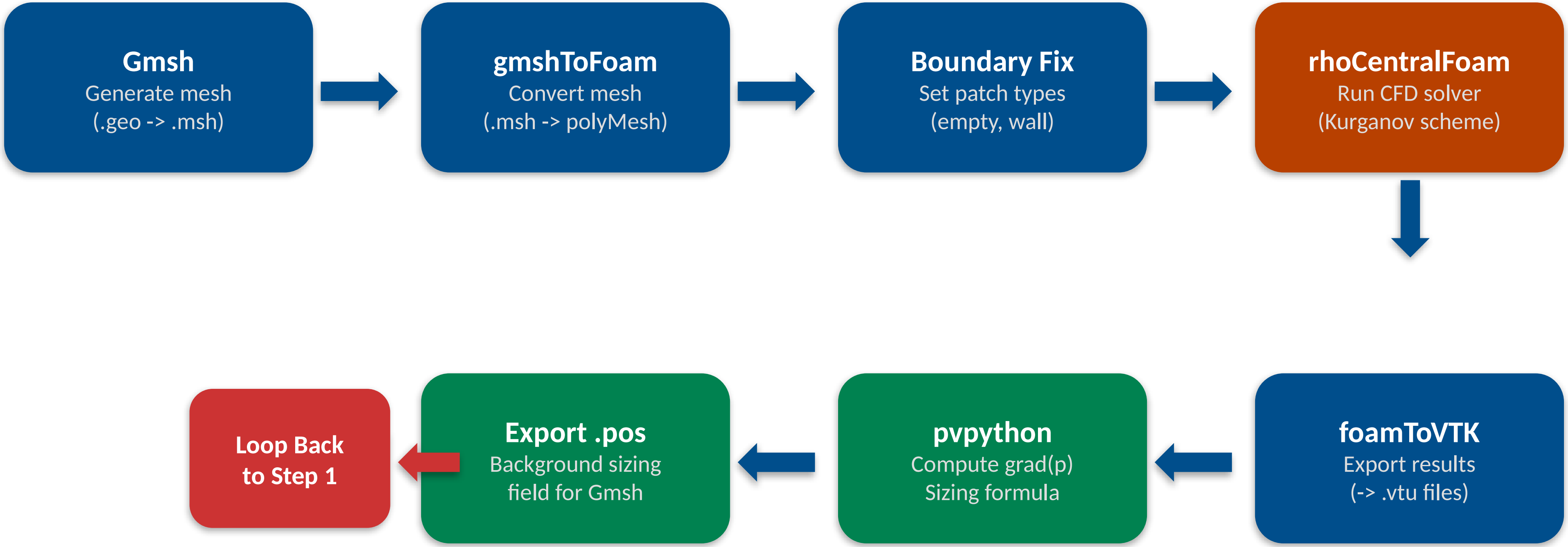
Parameter	Value
Time integration	Euler (1st order)
Initial time step	1e-8 s
Max Courant number	0.3
Max time step	1e-5 s
End time	0.001 s
Write interval	0.0002 s

Why These Choices?

- rhoCentralFoam: Only OpenFOAM solver that handles Mach 3 on unstructured mesh
- sonicFoam crashes at ANY supersonic speed on triangular mesh
- Minmod: Most stable limiter; vanLeer/vanAlbada produce oscillations at Mach 3
- maxCo = 0.3: Required for stability with strong bow shock
- Euler time integration: Sufficient for steady-state convergence

AMR Pipeline Architecture

Automated Workflow: all_run.sh



Configuration: amr_pipeline.input controls everything: loops=3, extraction_mode=gradient, gradient_field=p, sizing_min=0.002, sizing_max=0.015, sizing_scale=100.0, solver=rhoCentralFoam. No script editing required.

Gradient-Based Mesh Sizing Formula

$$h(x) = h\_min + (h\_max - h\_min) / (1 + alpha * |grad(p)| / |grad(p)|\_max)$$

Frey & Alauzet (2005), Loseille & Alauzet (2011)

Symbol	Parameter	Value	Description
h_min	Minimum cell size	0.002 m	Smallest allowed element in refined regions
h_max	Maximum cell size	0.015 m	Largest allowed element in smooth regions
alpha	Scaling factor	100.0	Controls sensitivity to gradient magnitude
grad(p)	Gradient field	Pressure	Chosen adaptation criterion

How It Works

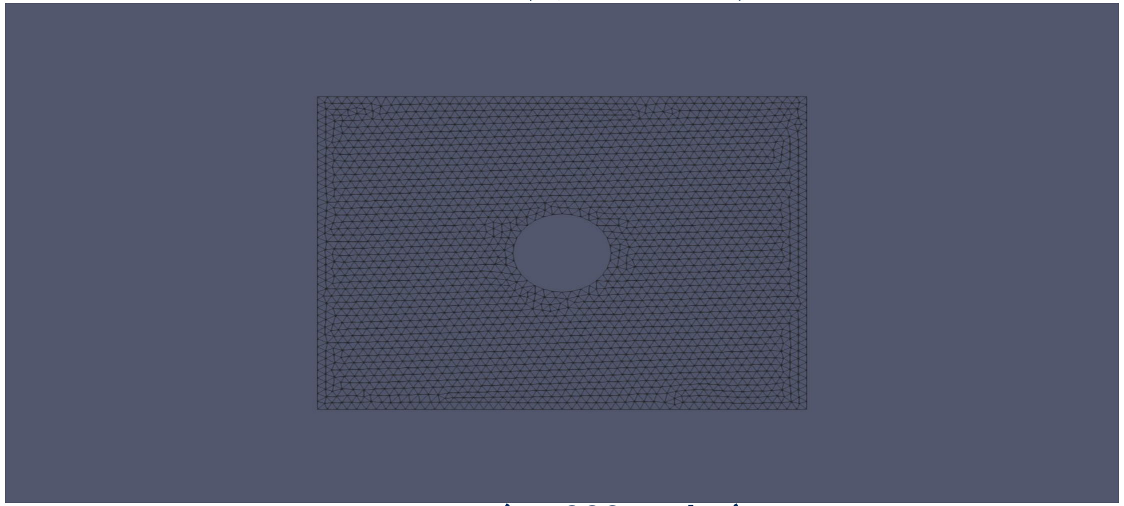
High pressure gradient (shock region): h approaches h_min = 0.002 m (fine cells concentrate at bow shock, wake, stagnation)

Low pressure gradient (smooth flow): h approaches h_max = 0.015 m (coarse cells save computation in freestream)

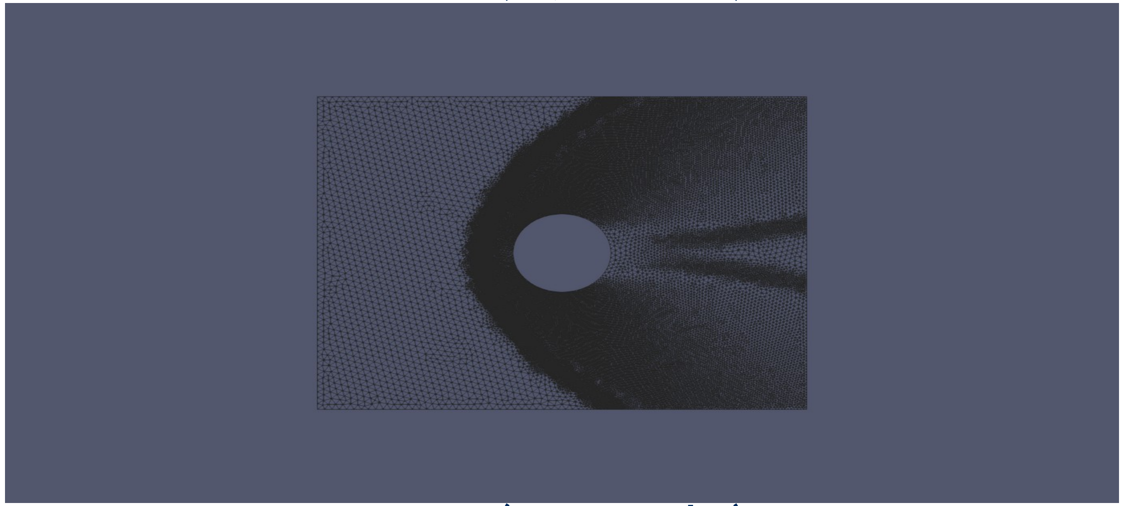
Mesh Progression

Uniform Mesh through 3 AMR Iterations

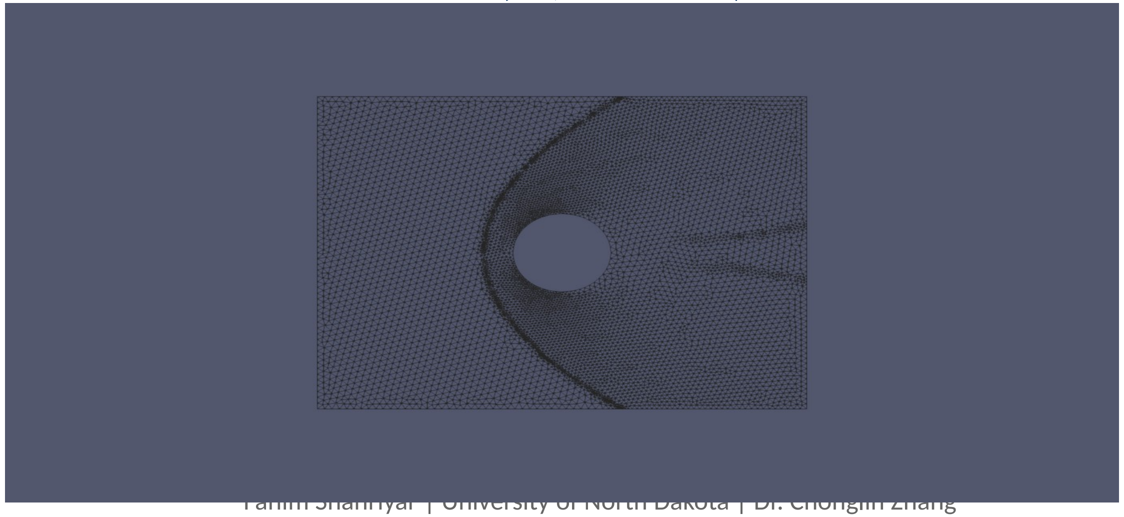
Uniform (4,804 nodes)



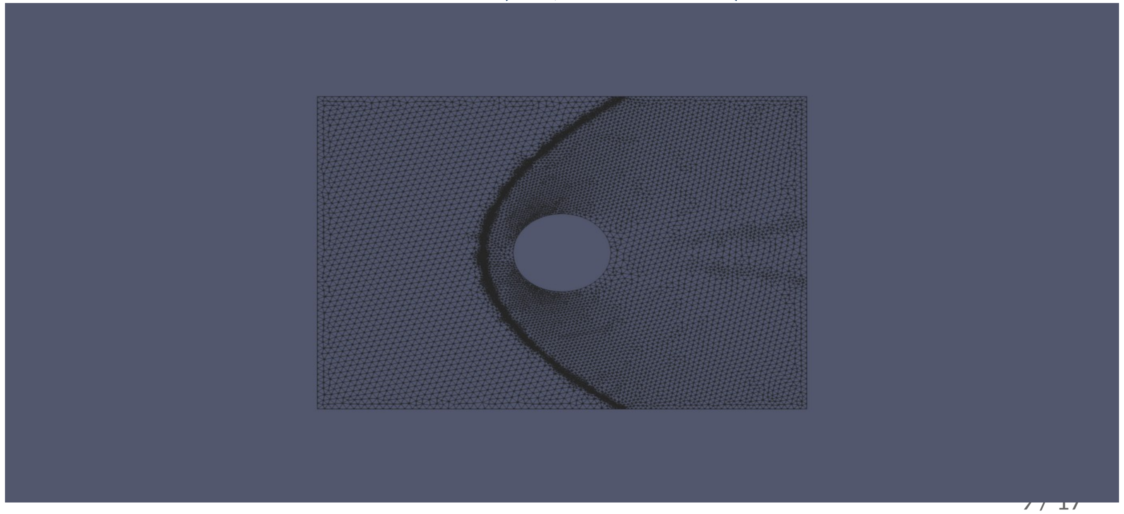
AMR 1 (71,904 nodes)



AMR 2 (14,888 nodes)



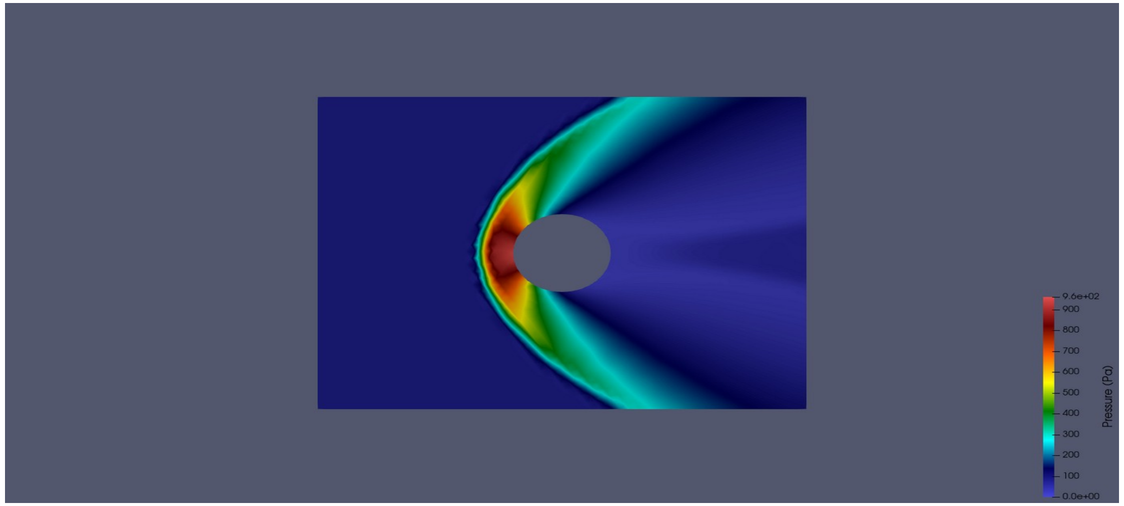
AMR 3 (17,500 nodes)



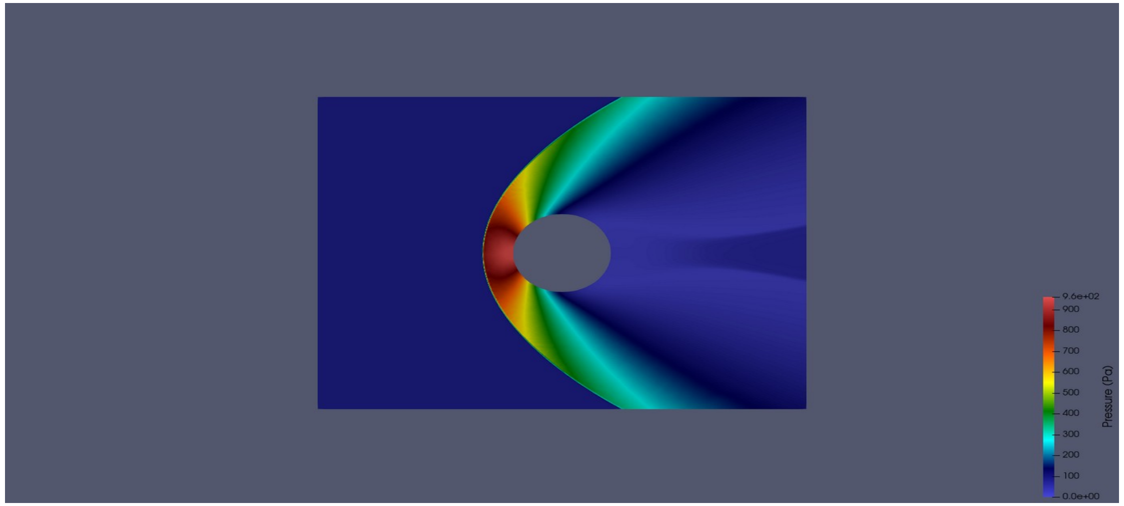
Pressure Field Comparison

Consistent scale: 0 - 960 Pa

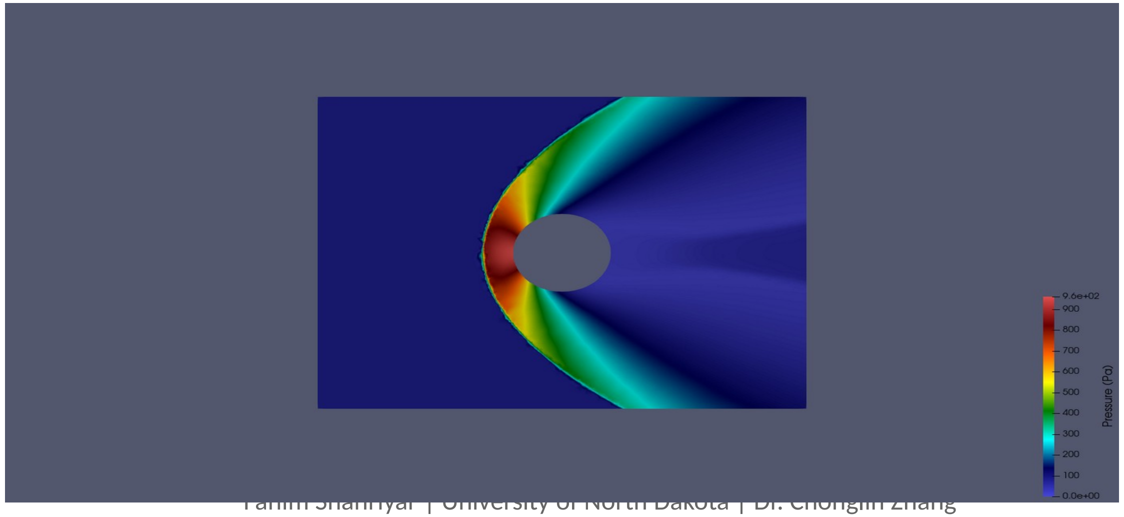
Uniform Mesh



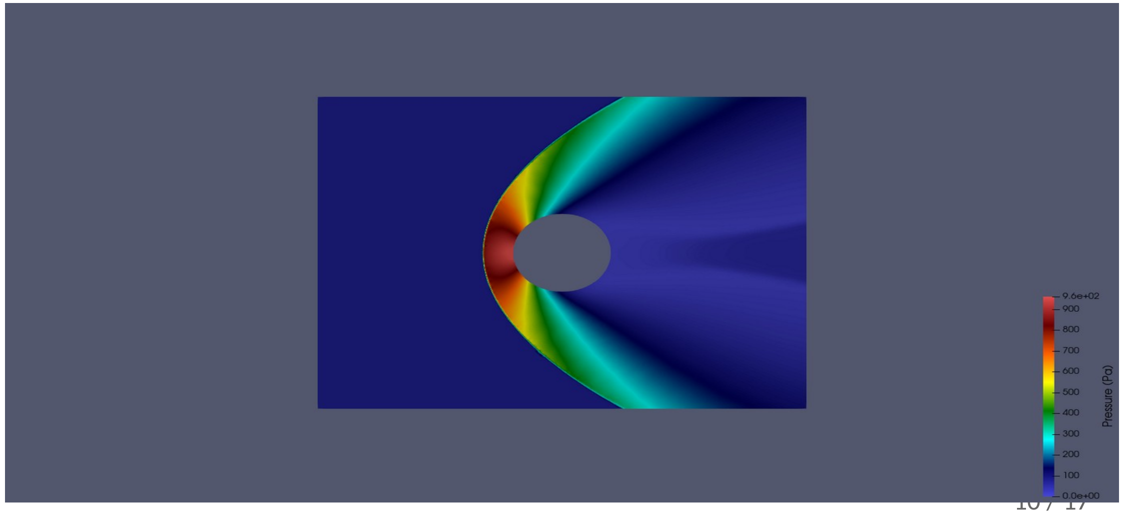
AMR 1



AMR 2



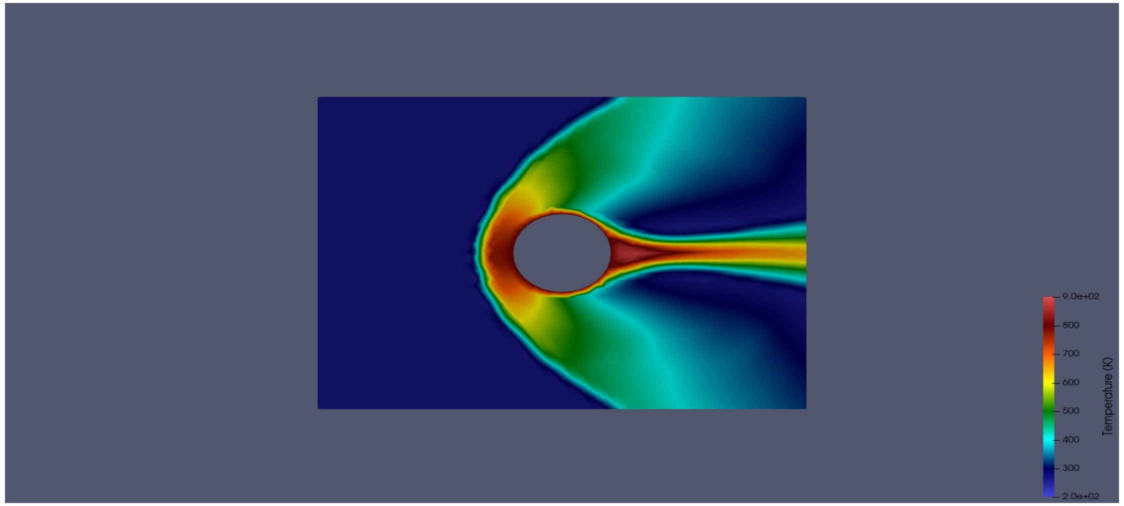
AMR 3



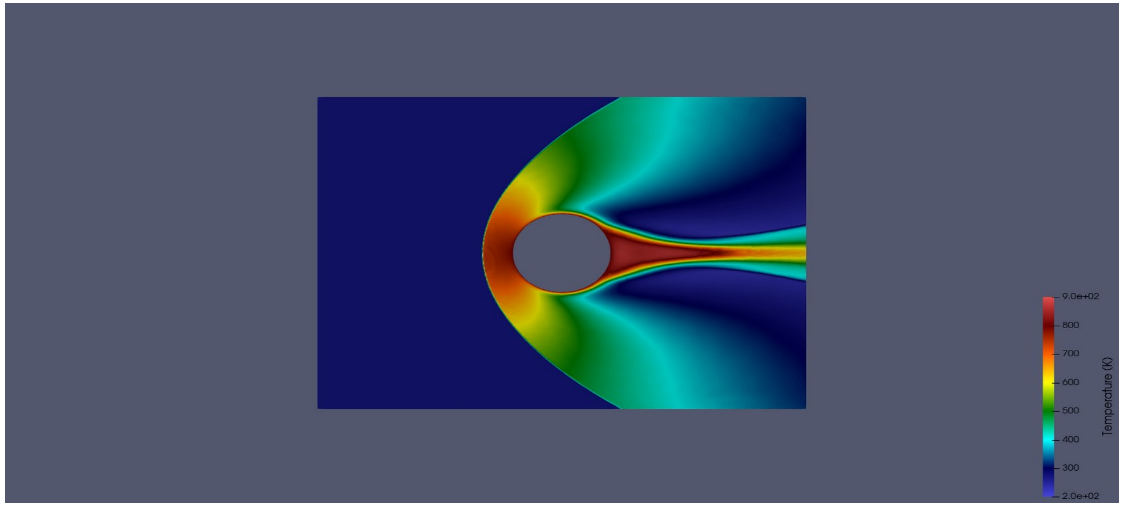
Temperature Field Comparison

Consistent scale: 200 - 900 K

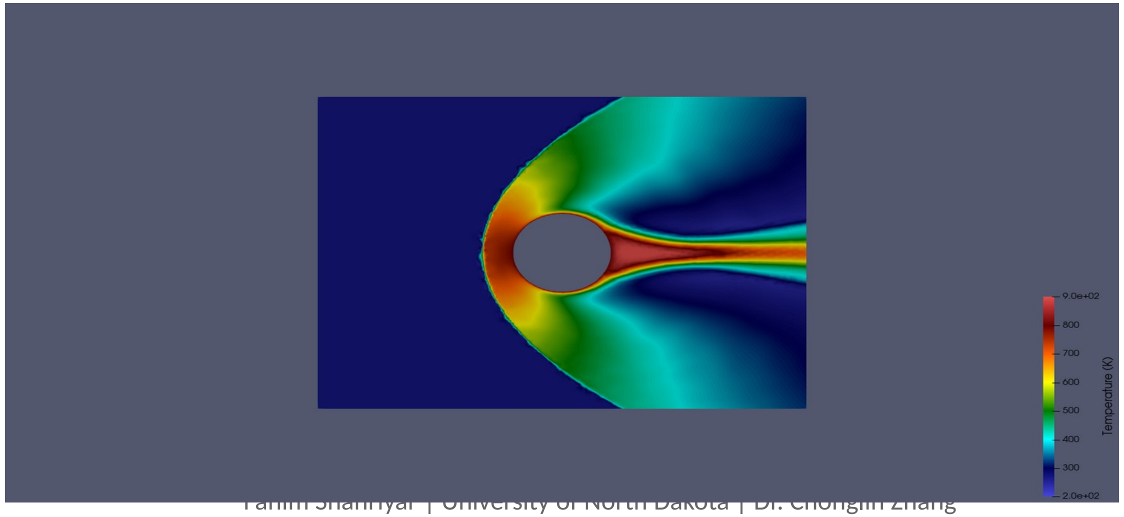
Uniform Mesh



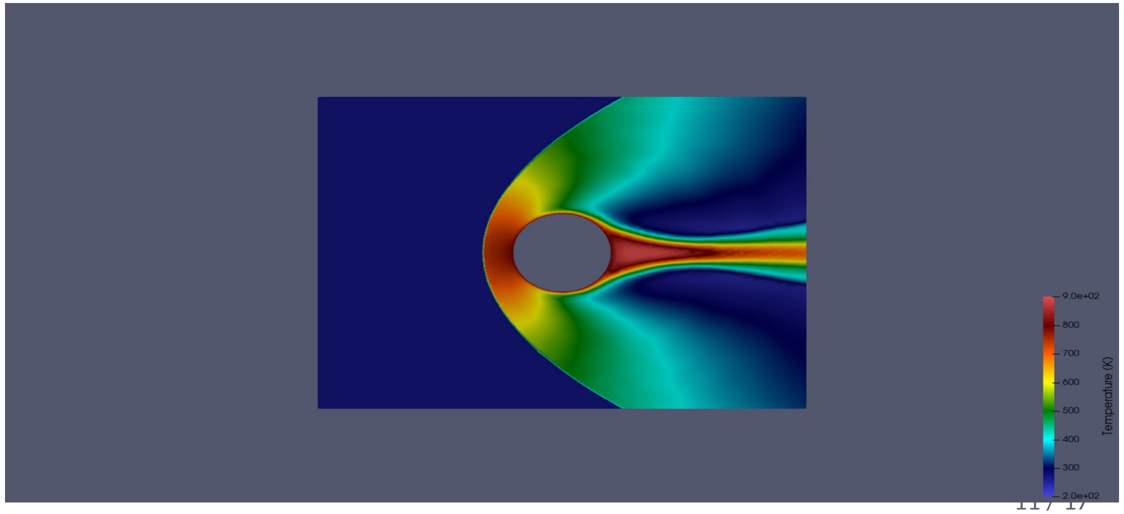
AMR 1



AMR 2



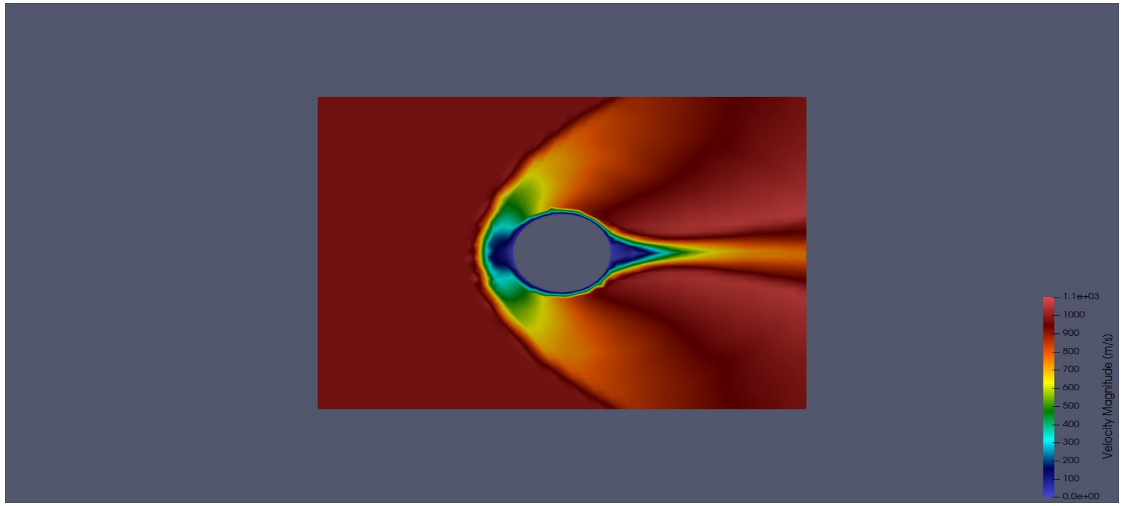
AMR 3



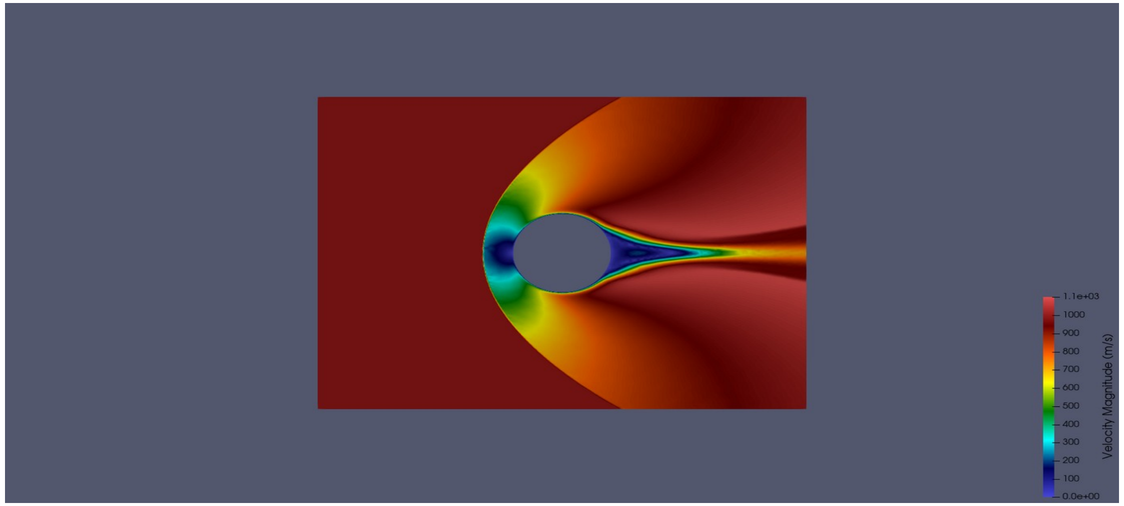
Velocity Magnitude Comparison

Consistent scale: 0 - 1100 m/s

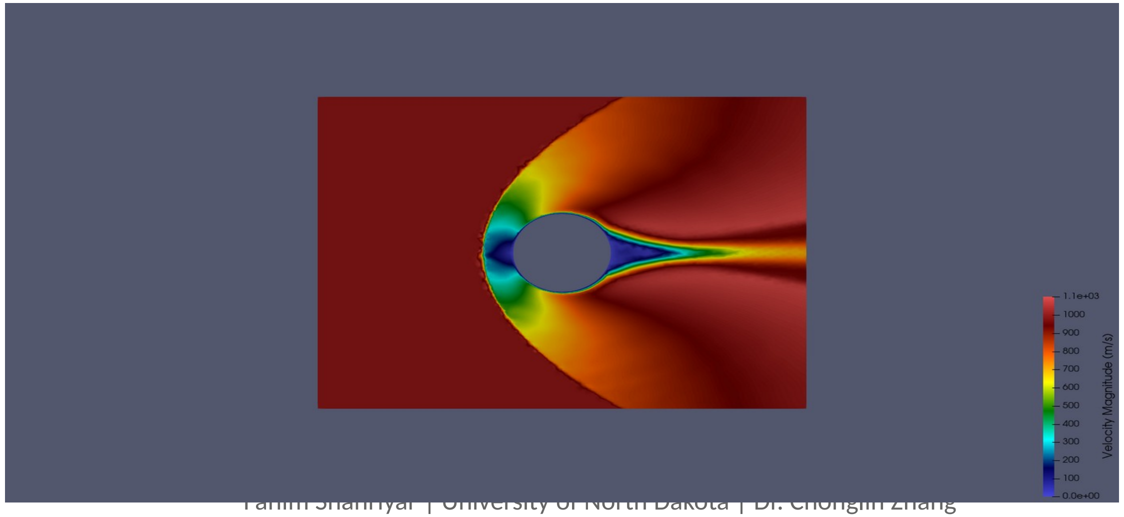
Uniform Mesh



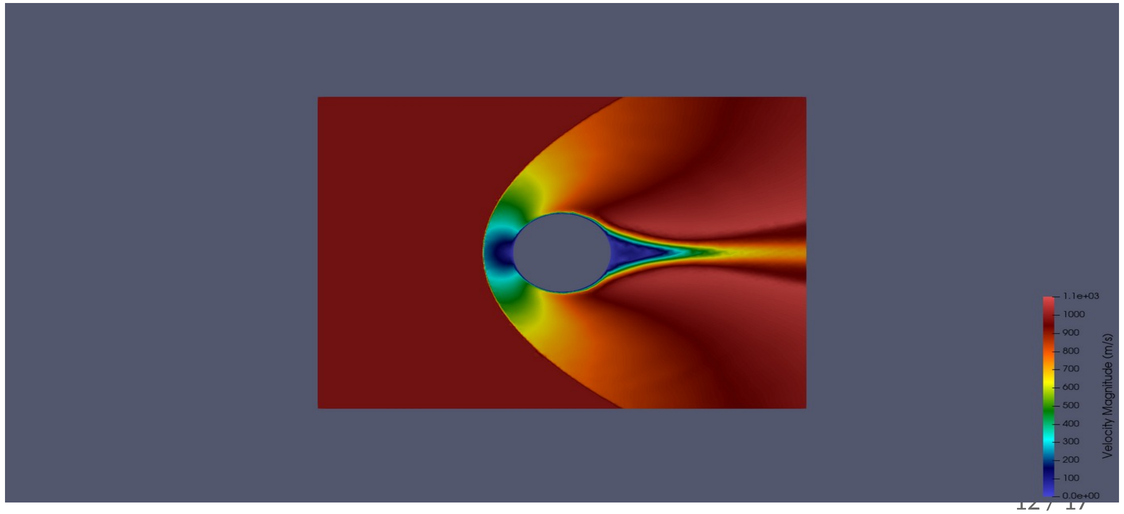
AMR 1



AMR 2



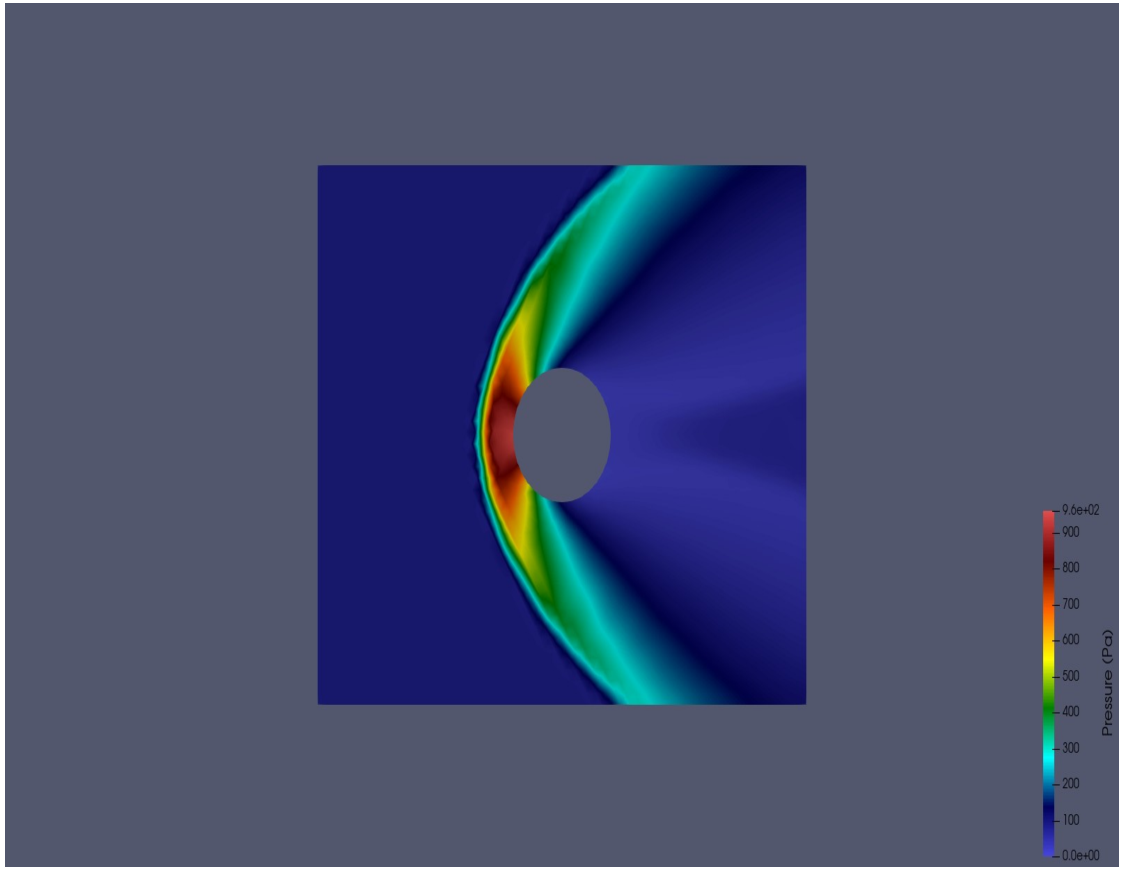
AMR 3



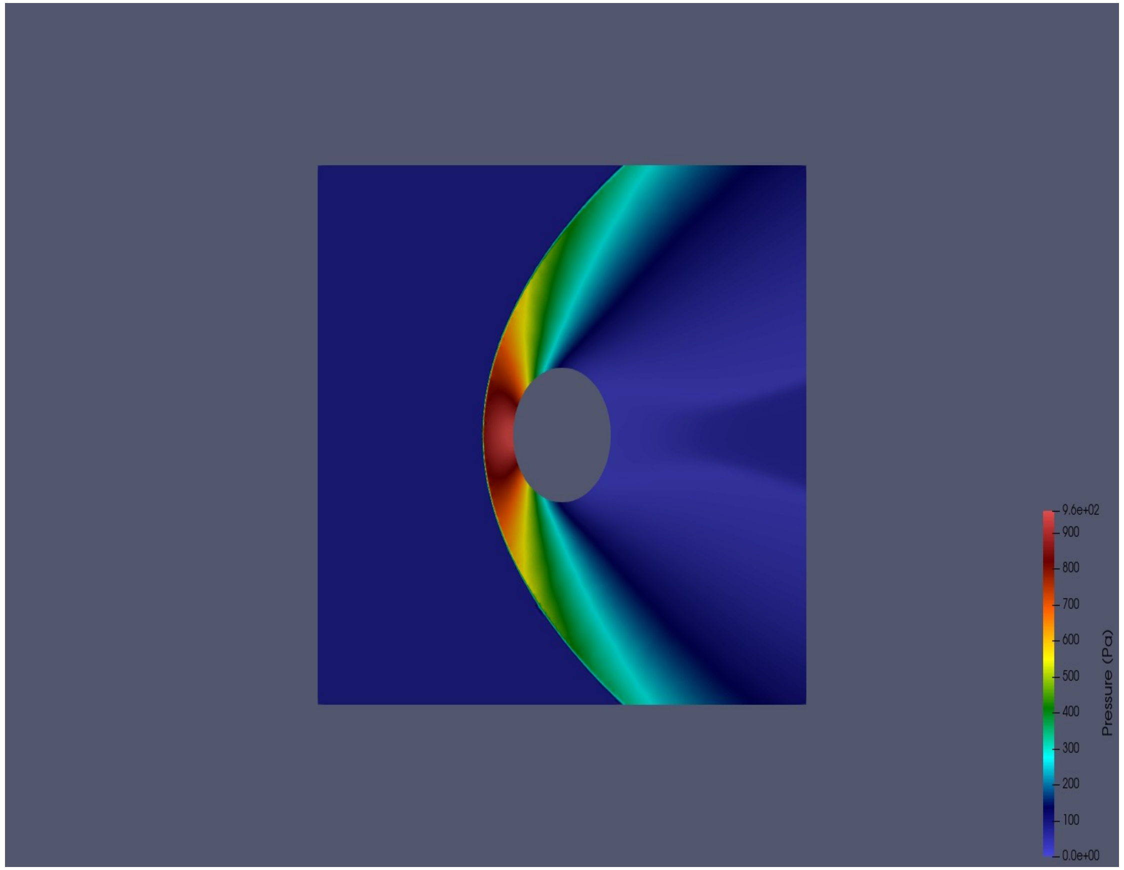
Pressure: Uniform vs AMR 3

Shock resolution improvement with gradient-based adaptation

Uniform (4,804 nodes)



AMR 3 (17,500 nodes)

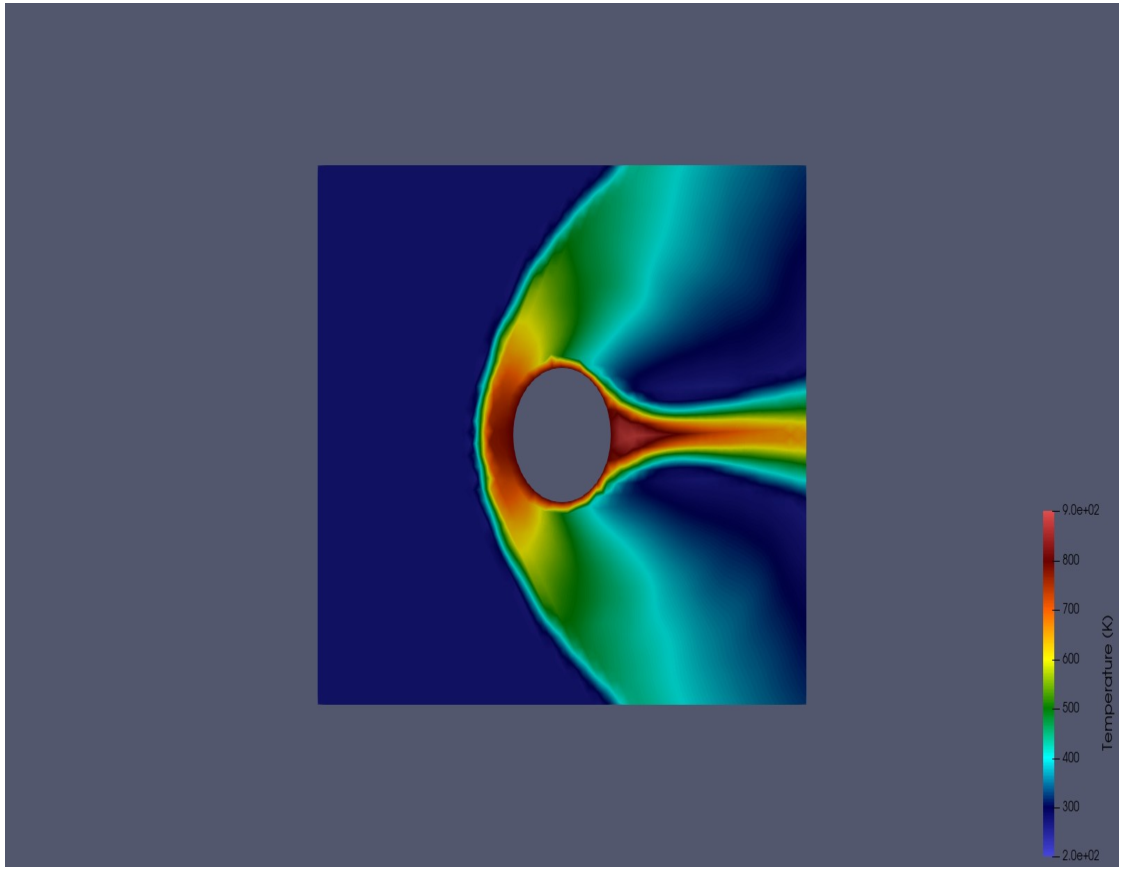


AMR 3 resolves the bow shock structure with sharper gradients using 3.6x more nodes concentrated at high-gradient regions

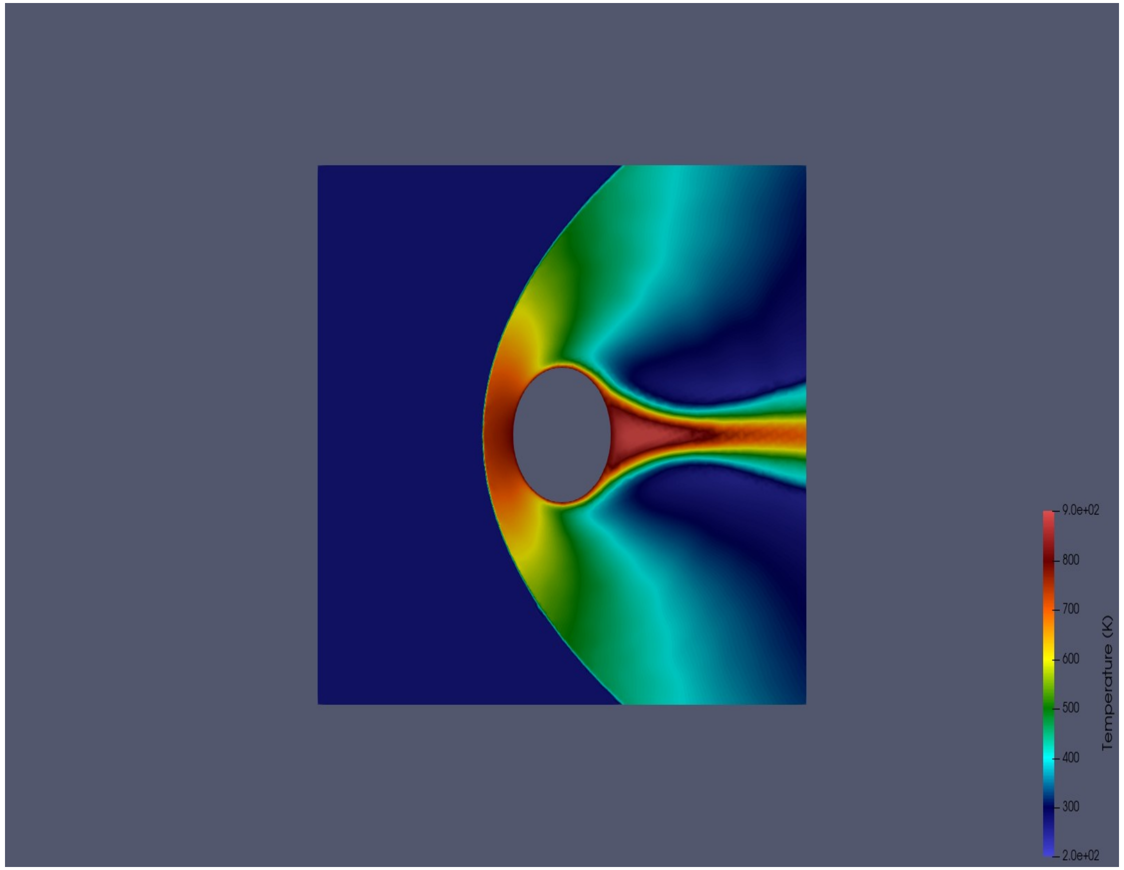
Temperature: Uniform vs AMR 3

Thermal structure near cylinder wall and wake

Uniform (4,804 nodes)



AMR 3 (17,500 nodes)



Temperature peaks at stagnation point (~900 K) due to adiabatic compression; AMR captures thermal layer thickness more accurately

Mesh & Results Summary

Mesh	Nodes	Cells	Mesh File Size	p Range (Pa)	T Range (K)
Uniform	4,804	4,592	652 KB	9.3 - 953.3	256.5 - 883.0
AMR 1	71,904	71,275	11 MB	11.8 - 973.7	233.0 - 864.0
AMR 2	14,888	14,528	2.2 MB	12.4 - 961.5	237.6 - 912.7
AMR 3	17,500	17,141	2.6 MB	12.6 - 970.9	241.0 - 907.9

Observations

AMR 1 over-refines (71k nodes) because initial gradients are everywhere

AMR 2 corrects by re-solving on AMR 1 mesh; now gradients are well-resolved

AMR 3 adds refinement only where residual gradients exist (17.5k nodes)

Final mesh (AMR 3) is 3.6x larger than uniform but 4.1x smaller than AMR 1

Pressure/temperature ranges converge across iterations

AMR Pipeline Parameters

Adaptation criterion: Pressure gradient |grad(p)|

h_min = 0.002 m (finest cells at shock front)

h_max = 0.015 m (coarsest cells in freestream)

Scaling factor alpha = 100.0

Gmsh CharacteristicLengthMin = 0.002 (AMR geo)

Gmsh CharacteristicLengthMax = 0.015 (AMR geo)

3 AMR iterations (configurable in amr_pipeline.input)

Key Findings

Solver Performance

rhoCentralFoam successfully handles Mach 3 on unstructured Gmsh mesh

sonicFoam crashes at ANY supersonic speed on this mesh type

Minmod limiter is essential; other limiters cause oscillations at Mach 3

maxCo = 0.3 required for stability with strong bow shock

AMR Effectiveness

Pressure gradient captures bow shock, stagnation, and wake accurately

3 AMR loops sufficient for mesh convergence (17.5k vs 4.8k nodes)

Self-correcting: AMR 1 over-refines, AMR 2-3 converge to optimal

Fully automated via all_run.sh with single config file

Limitations Discovered

Mach 3.5+ crashes: negative temperature from shock oscillations

Mach 27+ (re-entry): needs hy2Foam (hyStrath) for thermochemical non-equilibrium

hyStrath incompatible with OpenFOAM v2406

Subsonic flows (Mach 0.5): zero gradients make AMR meaningless

Pipeline Features

Universal: works with both OpenFOAM (CFD) and COMET (DSMC)

Two extraction modes: gradient (OpenFOAM) and direct (COMET MFP)

Solver-agnostic: reads VTU files, not solver-specific formats

Open-source ready: users configure ONLY amr_pipeline.input

Conclusions & Future Work

Conclusions

Successfully demonstrated gradient-based AMR for Mach 3 supersonic cylinder

Pressure gradient is an effective adaptation criterion for shock-dominated flows

The automated pipeline (Gmsh + OpenFOAM + ParaView) reduces manual meshing effort

AMR achieves better shock resolution than uniform mesh with moderate cell count increase

The pipeline is configurable, solver-agnostic, and ready for other flow problems

Future Work

Apply pipeline to 3D DSMC (COMET) results using direct MFP extraction

Test with friend's RAMC-II (Mach 27) DSMC results when available

Investigate multi-criteria adaptation (combine grad(p) + grad(T))

Extend to higher Mach numbers with structured/hybrid meshes

Integrate pipeline into thesis as AMR chapter demonstration

Thank You

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